

SNAPBURST

Governed autonomy for unmanned aircraft

When communications or command authority become uncertain, SnapBurst keeps aircraft behaviour authorised, bounded and auditable.

The envelope is pre-authorised. The drone self-governs inside it. The evidence proves it afterward.

Powered by the SnapSpace OS governance
core

SnapBurst

Current validation boundary: PX4 and ROS 2 software-in-the-loop.

Autonomous drones fail trust when command authority disappears

BVLOS aircraft cannot assume continuous command-and-control availability.

Lost-link safety is necessary, but blunt.

Conventional responses protect the aircraft by holding position, returning home, landing, aborting, or waiting for operator intervention.

That keeps the aircraft safe, but it does not scale to useful autonomous operations.

The hard problem is authority loss.

Who may command? What is permitted? How is the decision evidenced afterward?

Telemetry loss matters. Governance failure is the operational blocker.

C2 gaps should not end the mission.

A drone-specific governance and assurance layer

SnapBurst operates between mission autonomy and aircraft-command execution.

Autonomy proposes. SnapBurst governs. PX4 executes permitted actions.

PERMIT

dispatch permitted
action

REJECT

block unauthorised
action

HOLD

contain for escalation
or evidence

SnapBurst preserves

- command origin
- aircraft identity
- current authority
- governance verdict
- state transition
- aircraft response
- execution evidence

It does not replace the airframe, autopilot, mission planner, communications system, perception or detect-and-avoid system.

SnapBurst is the product.
SnapSpace OS is the proprietary governance core beneath it.

Three operating modes

One assurance layer across the aircraft lifecycle.

1

Manual Flight

**Human commands.
SnapBurst observes.
PX4 executes.**

The human operator keeps control priority through the C2 link. SnapBurst records command source, authority state, aircraft identity and outcome into the audit.

2

FreeFlight Recovery

**C2 loss.
Bounded continuation.
Verified handback.**

The aircraft continues only inside its pre-authorised operating envelope until governed restoration and handback.

3

Autonomous Work Mode

**Operator supervises.
Mission logic proposes.
SnapBurst governs.**

Mission logic proposes actions while C2 remains available for visibility and intervention.

Right mode. Right authority. Right evidence.

Communications loss becomes a governed state transition

FreeFlight is bounded mission continuation, not uncontrolled autonomy.

- 1 Envelope authorised before departure
- 2 Normal operation starts with C2 available
- 3 Identity, mission state and authority context are maintained
- 4 C2 interruption is detected
- 5 FreeFlight prevents unavailable authority from being treated as active command
- 6 Mission logic proposes only inside the authorised envelope
- 7 Actions are permitted, rejected or held
- 8 Commands outside the boundary do not reach PX4
- 9 External command is restored only by governed handback
- 10 Decision and execution sequence is preserved as evidence

The link disappears. The mission remains governed.

Ten aircraft, independently governed

Multiple aircraft lanes within one structured demonstration.

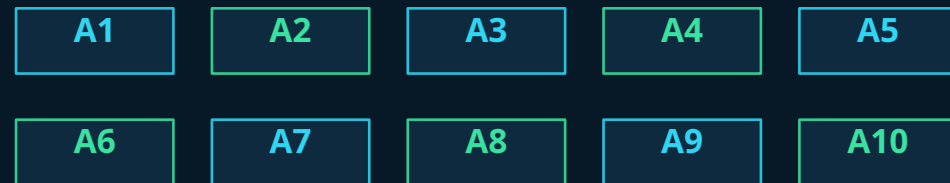
The current SnapBurst demonstration exercised ten independent PX4 SITL aircraft lanes.

Each lane maintained its own:

- identity
- mission state
- command source
- governance verdict
- dispatch path
- recovery state
- execution evidence

Demonstration phases:

- single-aircraft C2 loss
- five-aircraft C2 loss
- all-ten-aircraft C2 loss
- restoration and governed handback



This is independent multi-aircraft governance. It is not coordinated swarm mission behaviour.

N=10 C2-loss and restoration proof

One demonstration. Ten independently governed aircraft lanes.

7 / 7

phases passed

16 / 16

FreeFlight transitions

16 / 16

governed handbacks

0

foreign events

0

lane leakage

0

safety / stale
ACK events

Demonstrated capability

- maintain separate authority for multiple aircraft lanes
- enter bounded FreeFlight under simulated C2 loss
- preserve lane-specific attribution
- restore external command through governed handback
- prevent lane-command or evidence contamination

Boundary

Single PX4/ROS 2 SITL demonstration. Does not establish real-radio performance, physical flight, aircraft-to-aircraft coordination, leadership transfer, task reassignment or distributed swarm control.

Wider current validation record

Separate execution campaigns - not combined totals.

Four-hour multi-scenario campaign

14,405 seconds
222 / 222 tests passed
8 scenario groups
7 full cycles + 1 time-limited partial

Single-host sequential SITL, not simultaneous N=10 endurance.

Governed FreeFlight scale ladder

9 / 9 rungs passed
N=1, N=2, N=3, N=5 and N=10
0 recorded cross-namespace leakage
0 recorded stale artefacts

Governance and routing of FreeFlight commands, not trajectory accuracy.

Dual-bridge isolation

7 / 7 tests passed
Sequential and concurrent verdict combinations
Correct state continuity
Correct topic attribution

N=2 isolation only; no dual-bridge recovery or N>2 coordination claim.

Real PX4 command acknowledgement

1 / 1 proof passed - governed SET_MODE dispatched - PX4 result ACCEPTED - command acknowledgement correlation confirmed.

SnapBurst adds governance without replacing the aircraft system

OEM systems provide the body. SnapBurst governs the action boundary.

Existing aircraft and OEM systems provide

- airframe and propulsion
- flight controller
- communications
- mission autonomy
- perception and DAA
- sensors and payloads

SnapBurst provides

- command-authority governance
- authorised envelope enforcement
- C2-loss handling
- bounded FreeFlight
- governed handback
- lane attribution
- decision evidence

Operating relationship

OEM sensors detect.
Mission autonomy proposes.
SnapBurst governs.
PX4 executes permitted actions.
Evidence records what occurred.

SnapBurst may govern an avoidance action proposed by an OEM detect-and-avoid system. It does not claim to detect the obstacle or calculate the avoidance path.

The missing governance and assurance layer

SnapBurst sits between autonomous decisions and physical aircraft actions.

The market already contains capable products for aircraft manufacture, flight control, mission autonomy, fleet management, perception, communications and payload integration.

PX4 / ArduPilot

flight execution and predefined failsafe behaviour

Shield AI Hivemind

AI-pilot and mission-autonomy software

AuterionOS

broad aircraft, application and fleet operating platform

FlytBase

commercial drone automation and fleet workflows

Elsight / Doodle Labs

resilient communications and connectivity

SnapBurst addresses

- controlled behaviour as command conditions change
- bounded command relationships
- controlled continuation during disruption
- governed restoration of command
- attributable operational evidence

Two commercial routes

Standalone SnapBurst product for operators and integrators | SnapBurstOS integration and licensing for aircraft, autonomy and fleet-platform companies

The partner builds the body. SnapBurst provides the governed brain.

Governed brain means authority control, C2-loss handling, recovery and evidence generation - not an LLM pilot.

Hardware or platform partner provides

- aircraft or HIL platform
- autopilot and flight-control interface
- communications system
- sensors or payloads
- platform engineering
- controlled test access

SnapBurst provides

- command-governance integration
- authorised-envelope enforcement
- C2-loss and FreeFlight handling
- governed recovery and handback
- lane attribution
- correlated evidence package

First bounded demonstration

1. Integrate with one aircraft or HIL platform
2. Run normal mission segment
3. Introduce controlled C2 interruption
4. Confirm bounded continuation
5. Restore communications
6. Confirm governed handback
7. Produce evidence package

Objective: add governed autonomy and reviewable evidence around the partner system - not replace the autopilot.

Current position and next proof stage

Current evidence is strong inside the PX4/ROS 2 software-in-the-loop boundary.

Current validated position

- governed command decisions before egress
- bounded FreeFlight under simulated C2 loss
- governed restoration of external command
- independent lane attribution and isolation
- N=10 multi-aircraft governance in SITL
- real correlated PX4 SET_MODE acknowledgement

Outside current claim boundary

- hardware-in-the-loop validation
- physical-flight validation
- certified BVLOS capability
- detect-and-avoid performance
- distributed swarm coordination
- coordinator-loss tolerance
- production-aircraft integration

Seeking

- PX4-compatible partners
- controlled test facilities
- BVLOS programme support
- regulatory support
- funding for HIL and controlled-flight validation

SnapBurst governs the mission when command authority becomes uncertain - and proves what happened afterward.